MEDIKIOSK CLINICAL PLATFORM DOCUMENTATION

TECHNICAL REFERENCES

Technical References

Last Analyzed: 2026-09-16 Commit:`f118085510bdf2fa44785ee3ca5aaf5b4875c209`

1. Regulatory, Data, and Interoperability Standards

1.1 HL7 FHIR R4 (Fast Healthcare Interoperability Resources)

- **Description**: Standard specification for electronic health record exchange. Used to map patient demographic, clinical complaint, AYUSH assessment, and consent data into an interoperable JSON bundle.
- **Why Relevant**: Enables seamless integration with hospital information systems (HIS) and National Health Authority (NHA) registries.
- **Where Used**: `backend/src/fhir.ts:L3-L157`, `backend/src/app.ts:L185-L193`.
- **Citation / Link**: HL7 FHIR Release 4 Standard

1.2 ABDM (Ayushman Bharat Digital Mission) Specs

- **Description**: Specifications for M1 (ABHA creation & Aadhaar OTP authentication), M2 (Care Context linking & FHIR transmission), and M3 (HIU consent flow).
- **Why Relevant**: Supports nationwide digital health ecosystem integration under NHA India.
- **Where Used**: `backend/src/services/abdm.ts:L1-L260`, `backend/src/routes/abdm.ts`.
- **Citation / Link**: ABDM Developer Portal

1.3 Digital Personal Data Protection (DPDP) Act 2023 & Right to Erasure

- **Description**: Indian data privacy framework mandating explicit patient consent logs and hard-deletion (Right to Erasure / consent revocation).
- **Why Relevant**: Ensures legal compliance for capturing health and demographic data in OPD kiosks.
- **Where Used**: `backend/src/app.ts:L250-L279`, `backend/src/routes/sessions.ts:L152-L163`.
- **Citation / Link**: DPDP Act 2023 Gazette Notification

1.4 CMS FY2026 ICD-10-CM Coding System

- **Description**: International Classification of Diseases, Tenth Revision, Clinical Modification used to code clinical diagnoses and traditional medicine bridges.
- **Why Relevant**: Maps clinical summaries and traditional Ayurvedic conditions to standardized billing/diagnostic codes.
- **Where Used**: `rag_corpus/raw/ayurveda_icd/README.md:L60-L64`, `backend/src/summary.ts:L153-L174`.
- **Citation / Link**: CMS ICD-10-CM Official Guidelines

2. Artificial Intelligence & Machine Learning Architectures

2.1 Google Gemini LLM API (Gemini 3.6 Flash / 3.5 Flash)

- **Description**: Multimodal foundation model for RAG reasoning, document entity extraction, clinical summary generation, ICD-10 coding suggestions, and Hindi clinical translation.

- **Why Relevant**: Primary cloud LLM provider powering semantic triage and clinical summary generation.
- **Where Used**: `backend/src/app.ts:L158-L160`, `backend/src/rag/ayushRag.ts:L165-L215`, `backend/src/summary.ts:L136-L174`, `backend/src/clinicalAnalysis.ts:L12-L55`.
- **Citation / Link**: Google Gemini API Documentation

2.2 MedGemma 1.5 (via Ollama Local Runtime)

- **Description**: Domain-adapted medical LLM (`medgemma1.5:latest`) executed via Ollama on local hardware.
- **Why Relevant**: Primary offline/local LLM for semantic red-flag detection and off-script Q&A before falling back to Gemini.
- **Where Used**: `backend/src/orchestration/medgemmaClient.ts:L1-L24`, `backend/src/services/ollamaBootstrap.ts:L1-L56`.
- **Citation / Link**: TODO: verify official paper citation for MedGemma 1.5.

2.3 Sarvam AI API Suite

- **Description**: Sovereign Indic AI models:
  - **Saaras v3**: Multilingual Speech-to-Text (ASR) supporting Hindi, Tamil, Telugu, Bengali, Marathi, etc.
  - **Bulbul v3**: Multilingual Text-to-Speech (TTS).
  - **Doc AI Job Digitise**: Document OCR & structure parsing.
  - **Mayura v1**: Translation model between Indic languages and English.
- **Why Relevant**: Primary server-side Indic speech recognition, voice synthesis, document digitization, and translation.
- **Where Used**: `backend/src/orchestration/sarvamClient.ts:L1-L119`, `backend/src/orchestration/router.ts:L22-L58`.
- **Citation / Link**: Sarvam AI Developer Documentation

2.4 Tesseract.js (v7.0.0)

- **Description**: Pure JavaScript OCR engine based on Tesseract C++ engine compiled to WebAssembly.
- **Why Relevant**: Client-side and offline fallback OCR engine for reading medical prescription images.
- **Where Used**: `backend/src/app.ts:L15`, `backend/src/app.ts:L627`, `frontend/package.json:L25`.
- **Citation / Link**: Tesseract.js GitHub Repository

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3. Traditional Ayurvedic Literature & Clinical Frameworks

3.1 Dashavidha Pariksha (10-Fold Clinical Assessment)

- **Description**: Classical Ayurvedic diagnostic framework evaluating Prakriti, Vikriti, Agni, Koshtha, Sara, Samhanana, Pramana, Satmya, Sattva, Ahara Shakti, Vyayama Shakti, and Vaya.
- **Why Relevant**: Forms the core structure for the 12-dimension AYUSH intake questionnaire and deterministic reasoning module.
- **Where Used**: `backend/src/ayushReasoning.ts:L1-L150`, `backend/src/rag/ayushRag.ts:L114-L126`.
- **Source**: Charaka Samhita, Vimanasthana Chapter 8 (Rogabhishagjitiya Vimana).

3.2 Indian Vedas and Classical Ayurveda Corpus

- **Description**: Comprehensive text corpus comprising Rig Veda, Sama Veda, Yajur Veda, Atharva Veda, Charaka Samhita, Sushruta Samhita, Rasa Jala Nidhi, and IRJAY research articles.
- **Why Relevant**: Source raw materials indexed into vector storage for RAG retrieval during Ayush clinical reasoning.
- **Where Used**: `rag_corpus/raw/vedas/README.md:L1-L151`, `backend/src/rag/retriever.ts:L78-L86`.
- **Citation / Link**:

```bibtex
@misc{vedas_ayurveda_corpus_2026,
  title = {Indian Vedas and Ayurveda Corpus},
  year = {2026},
  publisher = {HuggingFace}
}
```